



The efficacy and safety of transcutaneous auricular vagus nerve stimulation in patients with mild cognitive impairment: A double blinded randomized clinical trial

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ABSTRACT

Background: There are 9.9 million new cases of dementia in the world every year. Short-term conversion rate from mild cognitive impairment (MCI) to dementia is between 20% and 40%, but long-term in 5–10 years ranges from 60% to 100%. It is particularly important to prevent or prolong the development of MCI into dementia. Both auriculotherapy and vagus nerve stimulation are effective on improving cognitive functions. However, there is no double blinded randomized clinical trial to support the effectiveness of transcutaneous electrical stimulation of auricular acupoints in patients with MCI.

Methods: This randomized controlled trial involved patients with MCI, aged from 55 to 75 years old. Patients were randomly allocated to transcutaneous auricular vagus nerve stimulation (taVNS) group or sham taVNS group. In the taVNS group, two auricular acupoints were stimulated, including heart (concha, CO₁₅) and kidney (CO₁₀), which are in the distribution of vagus nerve. While in the sham taVNS group, two other auricular acupoints were stimulated, including elbow (scaphoid fossa, SF₃) and shoulder (SF_{4,5}), which are out of the distribution of vagus nerve. The primary outcome was the Montreal cognitive assessment-basic, MOCA-B. The secondary outcomes included auditory verbal learning test-HuaShan version (AVLT-H), shape trails test A&B (STT-A&B), animal fluence test (AFT), Boston naming test (BNT), Pittsburgh sleep quality index (PSQI), rapid eye movement sleep behavior disorder screening questionnaire (RBDSQ), Epworth sleepiness scale (ESS) and functional activities questionnaire (FAQ). These outcome measures were taken at baseline, 24 weeks later.

Results: After 24 weeks of intervention, the data of 52 patients were intended for analysis. After intervention, there was significant difference in the overall scores of MoCA-B between taVNS group and sham taVNS group ($p = 0.033 < 0.05$). In taVNS group, compared with before intervention, the overall scores of MOCA-B increased significantly after intervention ($p < 0.001$). As for N5 and N7, the two sub-indicators of AVLT-H, in taVNS group, compared with before intervention, both N5 and N7 increased significantly after intervention (both $ps < 0.001$). As for STTB, in taVNS group, compared with before intervention, STTB was significantly reduced after intervention ($p = 0.016$). For BNT, in taVNS group, compared with before intervention, BNT increased significantly after intervention ($p < 0.001$). In taVNS group, compared with before intervention, PSQI, RBDSQ, ESS and FAQ decreased significantly after intervention ($p = 0.002, 0.025, < 0.001, 0.006$ respectively). 1 patient with a history of tympanic membrane perforation in taVNS group was reported with mild adverse reactions which disappeared a week after termination of taVNS. The intervention of taVNS is effective on increasing the overall scores of MoCA-B, N5 and N7.

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Conclusion: The clinical trial demonstrated that taVNS can improve cognitive performance in patients with MCI. This inexpensive, effective and innovative method can be recommended as a therapy for more patients with MCI in the prevention or prolonging of its development into dementia, but it is still required to be further investigated.

Trial registration: <http://www.chictr.org.cn>. (ID: ChiCTR2000038868)

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1. Introduction

According to the World Alzheimer's Disease (AD) Report in 2015 [1], there were 9.9 million new cases of dementia in the world every year, and the global dementia-related treatment and care costs have increased from \$604 billion in 2010 to \$818 billion in 2015.

In 2019, Jia Jianping's team [2,3] conducted a large-scale epidemiological survey in China, which showed that the prevalence rate of dementia among the elderly over 65 years old in China was 5.6%. At present, the number of patients suffering from dementia in China has exceeded 8 million, and it may exceed 26 million by 2040. Dementia has become a serious threat to the mental health and quality of life of the elderly in China, and also adds a heavy economic burden to the society and families.

Mild cognitive impairment (MCI) refers to the pathological state of pre-dementia with the manifestation of the progressive decline of memory or other cognitive functions but without decline of activities of daily life [4].

Studies [5,6] have shown that the short-term conversion rate from MCI to dementia is 20%–40% and that the long-term conversion rate in 5–10 years ranges from 60% to 100%. In order to prevent the occurrence and aggravation of AD, it is particularly important to study the diagnosis and treatment of MCI.

Sperling et al. [7] reviewed many studies and found that there was no obvious effective drug treatment for AD. A variety of new drugs, anti-amyloid beta (A β) drugs and new anti-tau drugs, for the treatment of mild to moderate dementia have failed to work [8]. Therefore, MCI in the early stage of AD as a new target of treatment attracted attention. However, so far, researchers have carried out on a variety of drugs, including cholinesterase inhibitors, non-steroidal anti-inflammatory drugs, ginkgo biloba preparations, chitosan lecithin and Vitamin E, all of which have been proved to be basically ineffective [9].

It is worth noting that vagus nerve stimulation (VNS), one of the physical therapy techniques, can improve cognitive function. Sun et al. [10] have demonstrated that invasive vagus nerve stimulation (iVNS) surgery can improve the cognitive function of patients with epilepsy. Sjögren et al. [11] confirmed that iVNS has a positive effect on the cognition of patients with AD. Merrill et al. [12] reported that after application of iVNS to 17 patients with AD for 6 months, 70% of them had no further cognitive impairment.

There is a unique vagus nerve distribution area in the surface of the auricular concha. Transcutaneous auricular vagus nerve stimulation (taVNS) has been creatively developed [13], which overcomes the limitations of iVNS surgery. It has also been preliminarily observed that transcutaneous auricular vagus nerve stimulation (taVNS) can improve the cognitive dysfunction in patients with epilepsy [14] and mild to moderate depression [15]. Kwon et al. [16] confirmed the clinical efficacy of ear acupuncture in the treatment of cognitive impairment through systematic review.

These studies have led to the hypothesis that transcutaneous electrical stimulation of auricular acupoints might be an effective management intervention for MCI.

To test this hypothesis, a randomized controlled trial was conducted to confirm the efficacy of transcutaneous electrical stimulation of auricular acupoints within the distribution of vagus nerve (taVNS group) for treating MCI, by using stimulation of auricular acupoints at the scapha out of the distribution of vagus nerve (sham taVNS group) as the comparator. Patients with MCI were randomly allocated into two parallel groups with the ratio of 1:1, taVNS group or sham taVNS group.

2. Methods

2.1. Settings and subjects

This double blinded, randomized controlled trial was conducted at Institute of Acupuncture and Moxibustion, China Academy of Chinese Medical Sciences, Beijing; Guang'anmen Hospital, China Academy of Chinese Medical Sciences, Beijing; Xuanwu Hospital, Capital Medical University, Beijing. The study protocol (version V1.1, Sep 18th, 2020) complied with the declaration of Helsinki and was approved by the Institutional Ethics Committee of Institute of Acupuncture and Moxibustion, China Academy of Chinese Medical Sciences, Beijing, and registered in <http://www.chictr.org.cn> (Identifier: ChiCTR2000038868) on October 8th, 2020 before the first participant was recruited.

2.2. Inclusion and exclusion criteria

Patients with MCI were required to meet the following four inclusion criteria: (1) men or women aged from 55 to 75 years, (2) at least half a year of local residence at the time of investigation; (3) Han Nationality; (4) Right-handed. Those who had (1) nervous system diseases, including cerebrovascular disease, encephalitis, brain tumor, brain trauma, epilepsy, Parkinson's disease and other neurological diseases, which might lead to cognitive decline; or (2) metabolic diseases, including anemia, thyroid dysfunction, folic acid and Vitamin B12 deficiency, which might lead to cognitive decline; or (3) severe mental illness, such as severe depression, or history of gas poisoning; or (4) history of general anesthesia; or (5) dementia; or (6) acute or serious life-threatening illness; (7) severe vision, hearing or language problems and could not complete neuropsychological assessment; (8) the scores of HAMD [17] and HAMA [18] ≥ 7 were excluded from the study.

2.3. Diagnostic criteria

The diagnosis was based on Jak/Bondi's criteria [19]. Participants were diagnosed as MCI if one of the following three criteria was met: 1) he or she had an impaired score, defined as >1 standard deviation (SD) below the age-corrected normative mean, on both measures within at least one cognitive domain (i.e., memory, language, or speed/executive function); 2) he or she had one impaired score, defined as >1 SD below the age-corrected normative mean, in each of the three cognitive domains sampled; or 3) he or she had

a score on the Functional Assessment Questionnaire (FAQ) = 9 [20] indicating dependence on three or more daily activities.

2.4. Randomization and blinding

The stratified block randomization was used and the center was taken as the stratification factor. An independent designer was in charge of a random number table generated by SAS software for formulating the allocation sequence of the two types of clamps, which was provided in a sealed envelope. An acupuncturist unsealed the given envelope and allotted the respective clamp according to the instructions. Therefore, the acupuncturist did not know which clamp was allotted to the patient.

The taVNS in this study was similar to Rong 2014 et al. [21]. A pair of clamps with the same appearance were placed for both ears. The clamp was designed with three carbon-impregnated silicone tips, one of which acted as the common terminal end to support the posterior surface of the auricle, the other two tips designed to stimulate the two skin surface points, one in the concha, the other in the scapha. For the latter two tips, only one of the two tips was active in a single clamp. In the taVNS group, the upper electrode with red marks was linked with the transcutaneous electrical nerve stimulator (TENS) but the lower with blue marks was not actively linked with the TENS (Fig. 1).

The acupuncturist allotted the respective clamp according to the allocation sequence. Both the acupuncturist and the patient did not know which tip was active when they received the clamps. The procedure was designed to ensure that the acupuncturist and the patient did not know whether the patient was receiving taVNS or

sham taVNS. Therefore, both the acupuncturist and the patient were blinded to group allocation (taVNS vs sham taVNS, 1:1). Unblinding was permitted at the end of this trial.

2.5. Interventions

After screening, all patients were trained to apply taVNS or sham taVNS. All patients were informed that different types of treatment were enrolled in this study, and that these interventions might be associated with sensation of electrical stimulation and positive outcomes. Specifically, patients were trained on how to turn on/off the stimulator, apply the electrode to the concha and scapha, locate the stimulating auricular acupoints, increase the intensity and fill out the diary. All subsequent treatments were self-administered by patients at home. Patients were also instructed to complete a patient diary each day to write the exact time, date and describe any side effects corresponding with or temporarily related to treatment. The investigators checked all diary once per month via WeChat (an instant messaging app, allowing its users to send words, emojis, pics, calls and video calls one-on-one with contacts, <https://web.wechat.com/>) video to enhance compliance. All procedures and stimulation parameters performed in sham taVNS group were identical to those in taVNS group.

In taVNS group, a pair of auricular acupoints were stimulated, including heart (concha, CO₁₅) and kidney (CO₁₀), in the distribution of vagus nerve. While in the sham taVNS group, another pair of auricular acupoints were stimulated, including elbow (scaphoid fossa, SF₃) and shoulder (SF_{4,5}), out of the distribution of vagus nerve. The scaphoid fossa was the stimulated site as the sham



Fig. 1. Interventions of taVNS and sham taVNS.

control in this study, which was different from the stimulation of the ear lobe as the sham stimulation in some studies [22–24]. The interventions were illustrated in Fig. 1.

All of the acupuncturists had at least two-year acupuncture practice. Patients should be seated or lie on the back. Stimulation points were disinfected and attached with the ear clips. The SDZ-IIIB electronic stimulator (Suzhou Medical Supplies Factory) was applied. Parameters were set, including (1) a pulse train of 20 Hz for 10 s and 100 Hz for 50 s in each minute, (2) intensity from 0.6 mA to 1.0 mA, determined by individual tolerance of the patients, (3) 30 min in each session, two sessions every day, once in the morning, the other in the afternoon or evening, (4) five consecutive days per week with an interval of 2 days for rest, (5) treatment period for 24 weeks. In response to harms, participants' request, or worsening of the disease, the participant was withdrawn from the study and given corresponding medical care. Participants were asked to write diaries to record the process of using taVNS device to ensure compliance to corresponding intervention and give response by telephone or WeChat. For patients with MCI, they could keep regular lifestyles. Participation of other trials related to cognitive function was not permitted during the trial. Patients were connected via WeChat video by an independent investigator once per week during the treatment to ensure that he/she was stimulating the specified acupoints.

2.6. Outcome assessments

The primary outcome was the Montreal cognitive assessment-basic (MOCA-B) [25,26]. The secondary outcomes included auditory verbal learning test-HuaShan version (AVLT-H) [27], shape trail test (STT-A&B) [28], animal fluency test (AFT) [29], Boston naming test (BNT) [30], Pittsburgh sleep quality index (PSQI) [31], rapid eye movement sleep behavior disorder screening questionnaire (RBDSQ) [32], Epworth sleepiness scale (ESS) [33] and functional activities questionnaire (FAQ) [20]. These outcomes were assessed at baseline and 24 weeks after baseline. Modification of the registration of this trial was as follows: RBDSQ and ESS were added as additional secondary outcomes to evaluate how taVNS impact on sleep quality. HAMD and HAMA were removed from secondary outcomes but to exclude patients with depression and anxiety.

2.7. Informed consent

Informed written consent was obtained before the beginning of the baseline tests. Participants without identity was disclosed in order to protect confidentiality before, during and after the trial.

2.8. Sample size estimation and statistical analysis

In this study, the method of group design was used to estimate the sample size by comparing the means of two samples.

$$n = 2[(\mu_{\alpha} + \mu_{\beta})^2 \sigma^2] / \delta^2$$

The letter “n” is the number of samples needed for a group. μ_{α} is the standard normal deviation of significance level. μ_{β} is the corresponding standard normal deviation of β . σ is the estimated standard deviation. δ is the difference between the mean values of two groups of numerical variables. α is taken as 0.05. β is set to 0.1. The power is 0.9 calculated by formula 1 minus β . Then for two-sided test, $\mu_{\alpha} = \mu_{0.05} = 1.96$, and $\mu_{\beta} = \mu_{0.1} = 1.28$. By referring to the previous literature [34] on auricular point therapy in the treatment of MCI, we estimated that in the treatment group MOCA-B score could be increased by 2.0 points, with a SD of 1.79 points

and that in the control group MOCA-B score could be decreased by 0.1 points, with a SD of 2.08 points. $\sigma = 1.79$, $\delta = 2.1$.

Sample size was estimated as follows:

$$n_1 = n_2 = 2[(1.96 + 1.28) \times 1.79]^2 / 2.1^2 \approx 24$$

Therefore, each group needed at least 24 cases for observation. According to the loss rate of 20%, each group needed at least 30 cases (24/0.8).

Patients with MCI were recruited through ads and notice near the three hospitals and WeChat. Two staff, rather than the investigators, were available to the data and recorded it into the file to ensure the data quality during data arrangement. The coordinating center was responsible for overseeing the trial.

The outcome analysis was carried out on basis of an intention-to-treat data set, defined as the subset of participants who completed at least one session of intervention, baseline and at least one post-baseline evaluation. The primary outcome was MOCA-B. The secondary outcomes included AVLT-H, STT-A&B, AFT, BNT, PSQI, RBDSQ and ESS. Demographics were analyzed through independent sample *t*-test at baseline to determine equality of the two groups, namely taVNS group and sham taVNS group. Chi-square tests were used to evaluate the difference of the following categorical variables, including sex, social support, siblings alive, close friends, vocation, job category, marital status, person living with, hypertension (with or without), diabetes (with or without), hyperlipidemia (with or without), AD family history (with or without), habits of frequent drinking tea and coffee and having fish. Furthermore, Chi-square tests were also for hypertension (including regular administration and stable pressure), diabetes (including regular administration and stable blood glucose), hyperlipidemia (regular administration), AD family history (both relatives at the first degree and at the second degree).

Statistical analysis was performed via SPSS, version 15.0 (SPSS, Inc., Chicago). Normally distributed data were analyzed by with two-tailed *t*-tests. Categorical variables were analyzed with Chi-square tests. The significance level was set at a 2-sided $P < 0.05$. As for MOCA-B, AVLT-H, STT-A&B, AFT, BNT, PSQI, RBDSQ, ESS and FAQ, the factors were time (2 levels: before intervention, after intervention), groups (2 levels: taVNS, sham taVNS) and their interaction. Subgroup analyses were conducted to assess the effect of taVNS on the three cognitive domains including memory, execution and language.

2.9. Safety assessment

Participants were asked to report adverse events, such as unexpected physical change or side effects, by telephone or WeChat at any time during the study. Every reported adverse event was written in the case report. If the adverse event was severe and related to the trial, the participant was required to withdraw from the study and receive corresponding medical care.

2.10. Data safety monitoring

Data safety monitoring board members met every 24 weeks or at any time required. Participants who reported worsening of symptoms were withdrawn from the study and referred to proper treatment immediately. There were four gradings for safety.

Grading 1: secure, no adverse effects reported.

Grading 2: secure, mild adverse effects reported, no need for any treatment.

Grading 3: secure problems, moderate adverse effects reported, corresponding treatment required, continuing of intervention.

Grading 4: severe problems, suspension of intervention.

3. Results

Sixty patients were included in the study. Thirty of them were randomly assigned to taVNS group and the other thirty to sham taVNS group. All subjects (60) were included in intention-to-treat (ITT) population. 52 subjects (86.7%) were included in per protocol (PP) population. 25 patients in taVNS group and 27 patients in sham taVNS group were analyzed for the primary and secondary outcomes.

Eligible participants were recruited from Jan. 2021 to Oct. 2021.

3.1. The integrity of blinding

Throughout the study, no acupuncturist or patient could guess out whether the participant received taVNS or sham taVNS.

3.2. Baseline characteristics

The Consolidated Standards of Reporting Trials (CONSORT) flowchart was shown in Fig. 2. 303 participants with complaint of cognitive decline were screened in this study. 228 participants

were excluded from this trial according to the inclusion and exclusion criterion. 12 were left-handed. 46 were without the scope of 55–75 years old. 22 did not belong to Han nationality. 13 were not local Beijing residence. 135 did not meet the diagnostic criteria. Table 1 showed the baseline demographic and clinical characteristics for participants in each group. The participants in taVNS group and sham taVNS were matched on age, sex, education, social support, siblings alive, close friends, vocation, job category, marital status, person living with, hypertension (with or without), diabetes (with or without), hyperlipidemia (with or without), AD family history (with or without), habits of frequent drinking tea and coffee and having fish (all p s > .05). Furthermore, for hypertension, all of the following factors, including hypertension history, usual mean arterial pressure, regular administration and stable pressure, were matched (all p s > .05). For diabetes, the following factors were also matched, including diabetes history, regular administration and stable blood glucose (all p s > .05). For hyperlipidemia, both hyperlipidemia history and regular administration were matched (all p s > .05). As for AD family history, both relatives at the first degree and at the second degree were matched (all p s > .05). There was only one patient with coronary heart disease in taVNS group,

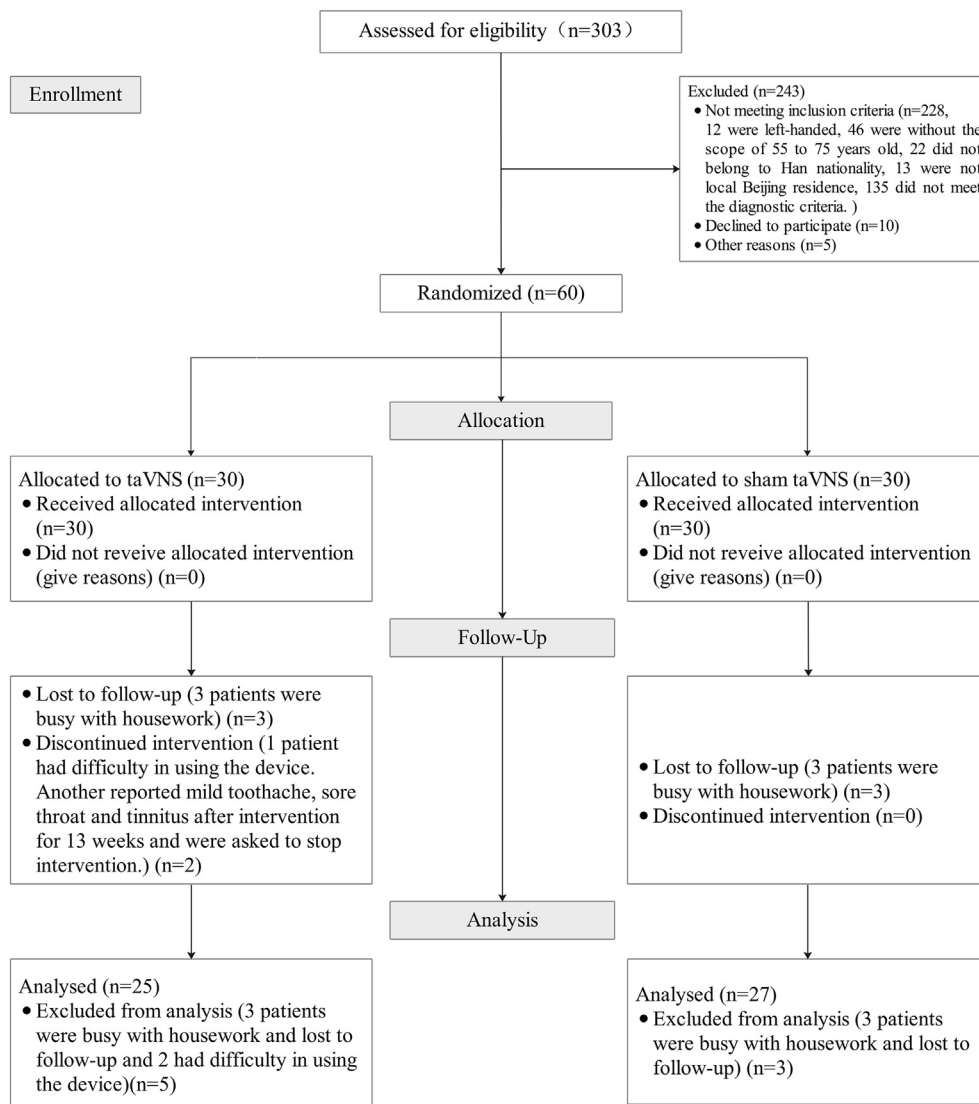


Fig. 2. The CONSORT flowchart.

Table 1
Baseline demographic and clinical characteristics of the participants.

Variables	taVNS (n = 25)	Sham taVNS (n = 27)	Statistics		
			t/ χ^2 /U	df	p
Age, mean (SD), year	66.9 (3.66)	67.0 (4.36)	−0.0383	50.0	0.970
Women, %	20 (80.00)	23 (85.19)	0.244	1.0	0.621
BMI, mean (SD), %	26.4 (3.79)	23.6 (3.50)	2.7269	50.0	0.009
Education, median, year	9	9	325		0.805
Social support (2 levels, 1 ^a , 2 ^b)	7, 18	5, 22	0.657	1	0.417
Siblings alive (2 levels, 0 ^c , 1 ^d)	1, 24	0, 27	1.101	1	0.294
Close friends (4 levels, 1 ^e , 2 ^f , 3 ^g , 4 ^h)	5, 7, 2, 11	1, 2, 13, 11	13.454	3	0.004
Vocation (16 levels ⁱ)	1,14,0,1,1,2,2,1,1,1,1,0,0,0,0,0	2,13,0,1,0,1,1,0,0,1,1,3,1,1,1,1	11.0	14.0	0.688
Job category (three levels ^j)	6,11,8	9,7,11	1.89	2.0	0.389
Marital status (three levels ^k)	17,2,6	18,3,6	0.152	2.0	0.927
Person live with (five levels ^l)	7,11,4,0,3	5,14,5,1,2	1.93	4.0	0.749
Hypertension (two levels, 0 ^m , 1 ⁿ)	18,7	18,9	0.173	1.0	0.677
Hypertension history (mean, S.D.)	14.1, 7.52	9.28, 6.69	1.368	14.0	0.193
Highest mean arterial pressure (mean, SD, median)	124.0, 12.37, 127.0	111.78, 8.17, 110.00	2.382	14.0	0.032
Usual mean arterial pressure	93.4, 7.59, 96	90.78, 7.56, 90	0.694	14.0	0.499
Regular administration (2 levels, 0 ^m , 1 ⁿ)	0, 7	2, 7	1.78	1.0	0.182
Stable pressure (2 levels, 0 ^m , 1 ⁿ)	0, 7	2, 7	1.78	1.0	0.182
Diabetes (two level, 0 ^m , 1 ⁿ)	22, 3	21, 6	0.948	1.0	0.330
Diabetes history (mean, S.D., median)	12.3, 9.29, 15.0	12.5, 10.49, 7.5	−0.0234	7.0	0.982
Regular administration (2 levels, 0 ^m , 1 ⁿ)	0, 3	1, 5	0.563	1.0	0.453
Stable blood glucose (2 levels, 0 ^m , 1 ⁿ)	2, 1	2, 4	0.900	1.0	0.343
Hyperlipidemia (two levels, 0 ^m , 1 ⁿ)	15, 10	11, 16	1.93	1.0	0.165
Hyperlipidemia history (median)	7.00	5.50	78.5		0.958
Regular administration (2 levels, 0 ^m , 1 ⁿ)	2, 8	2, 14	0.266	1.0	0.606
AD family history (two levels, 0 ^m , 1 ⁿ)	20, 5	25, 2	1.77	1.0	0.184
Relatives at the first degree (three levels, 0, 1, 2)	1, 4, 0	1, 0, 1	4.55	2.0	0.103
Relatives at the second degree (two levels, 0, 1)	4, 1	1, 1	0.630	1.0	0.427
Tea (five levels ^o)	12, 0, 2, 1, 10	9, 6, 2, 3, 7	7.89	4.0	0.096
Coffee (five levels ^o)	2, 1, 2, 3, 17	0, 4, 3, 6, 14	5.22	4.0	0.265
Fish (four levels ^o)	1, 2, 11, 11	1, 4, 9, 13	0.958	3.0	0.811

Note: ^a live alone, ^b live with family members, ^c no siblings alive, ^d siblings alive, ^e none, ^f one or two, ^g three to five, ^h 6 or more, ⁱ 16 levels included cadre, worker, farmer, individual business personnel, doctor, accountant, driver, salesperson, chef, scientific research personnel, freelancer, teacher, pharmaceutical personnel, pharmacist, service practitioner and cashier in sequence, ^j mental worker, manual worker, mental-manual worker in order, ^k married(spouse alive), divorced, widowed in sequence, ^l none, spouse, sons and daughters (SD), parents, spouse and SD. ^m without, ⁿ with, ^o every day, several times a week, once a week, very seldom, never in sequential order.

and no patient with coronary heart disease in sham taVNS group. The scores of HAMD and HAMS were less than 7 between patients in taVNS group and sham taVNS group.

3.3. Primary outcome

Before intervention, there was no significant difference in the overall scores of MoCA-B between taVNS group and sham taVNS group ($p = 0.285$). However, after intervention, there was significant difference in the overall scores of MoCA-B between taVNS group and sham taVNS group ($p = 0.033 < 0.05$). In taVNS group, compared with those before intervention, the overall scores of MoCA-B increased significantly after intervention ($p < 0.001$). However, in sham taVNS group, compared with those before intervention, there was no significant difference in the overall scores of MoCA-B after intervention ($p = 0.338$). Compared with sham taVNS, there was also significant difference in the difference value of pre- and post-intervention in taVNS group ($p < 0.001$). The results were summarized in Fig. 3a.

3.4. Secondary outcomes

3.4.1. AVLT-H

AVLT-H was for evaluating the memory function. There were two indicators for AVLT-H, N5 for evaluating immediate recall, and N7 for delayed recall.

Referring to N5, there was no significant difference between taVNS group and sham taVNS group ($p = 0.963$) before intervention. After intervention, there was also no significant difference between taVNS group and sham taVNS group ($p = 0.290$). In taVNS

group, compared with before intervention, N5 was significantly increased after intervention ($p < 0.001$). However, in sham taVNS group, compared with before intervention, there was no significant difference in N5 after intervention ($p = 0.059$). Compared with sham taVNS group, there was also significant difference in the difference value of pre- and post-intervention ($p = 0.047$) in taVNS group. Results were summarized in Fig. 3b.

As for N7, there was no significant difference between taVNS group and sham taVNS group ($p = 0.470$) before intervention. After intervention, there was also no significant difference in N7 between taVNS group and sham taVNS group ($p = 0.056$). In taVNS group, compared with before intervention, N7 was significantly increased after intervention ($p < 0.001$). However, in sham taVNS group, compared with before intervention, there was no significant difference in N7 after intervention ($p = 0.051$). Compared with sham taVNS, there was also significant difference in the difference value of pre- and post-intervention in taVNS group ($p = 0.005 < 0.01$). Results were summarized in Fig. 3c.

3.4.2. STTA and STTB

STTA and STTB were for evaluating the executive function.

With regard to STTA, there was no significant difference between taVNS group and sham taVNS group ($p = 0.999$) before intervention. After intervention, there was also no significant difference between taVNS group and sham taVNS group ($p = 0.667$). In taVNS group, compared with before intervention, STTA was not significantly different after intervention ($p = 0.538$). In sham taVNS group, compared with before intervention, there was also no significant difference after intervention ($p = 0.161$).

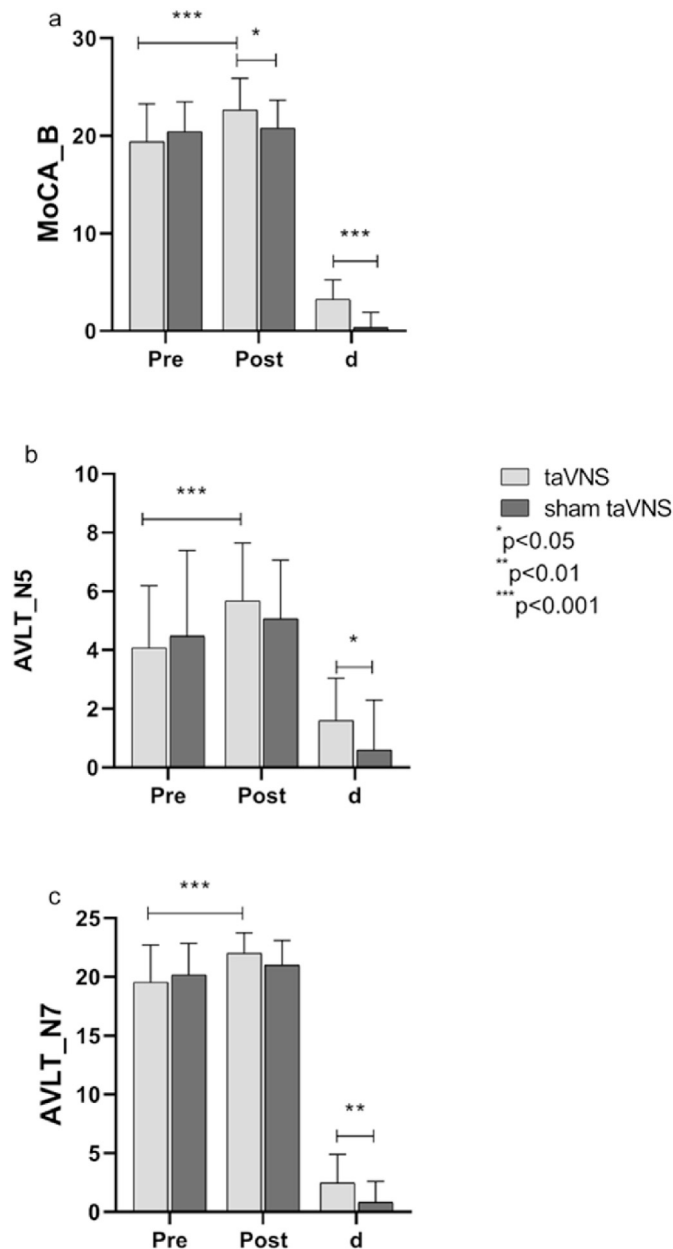


Fig. 3. Results of MoCA-B(a), AVLT_N5(b) and AVLT_N7(c). Note: Pre, before intervention; Post, after intervention; d, the difference value of before intervention and after intervention.

As for STTB, there was no significant difference between taVNS group and sham taVNS group ($p = 0.667$) before intervention. After intervention, there was also no significant difference between taVNS group and sham taVNS group ($p = 0.195$). However, in taVNS group, compared with before intervention, STTB was significantly different after intervention ($p = 0.016 < 0.05$). In sham taVNS group, compared with before intervention, there was also significant difference after intervention ($p < 0.001$).

3.4.3. AFT and BNT

AFT and BNT were for evaluating language functions.

About AFT, there was significant difference between taVNS group and sham taVNS group ($p = 0.010$) before intervention.

For BNT, there was no significant difference between taVNS group and sham taVNS group ($p = 0.632$) before intervention. After intervention, there was also no significant difference between

taVNS group and sham taVNS group ($p = 0.393$). In taVNS group, compared with before intervention, BNT was significantly increased after intervention ($p < 0.001$). Interestingly, in sham taVNS group, compared with before intervention, BNT was also significantly increased after intervention ($p < 0.001$).

3.4.4. PSQI, RBDSQ and ESS

There were three indices for evaluating sleep, including PSQI, RBDSQ and ESS.

In regard to PSQI, there was no significant difference between taVNS group and sham taVNS group ($p = 0.754$) before intervention. After intervention, there was also no significant difference between taVNS group and sham taVNS group ($p = 0.712$). In taVNS group, compared with before intervention, PSQI decreased significantly after intervention ($p = 0.002$). Interestingly, in sham taVNS group, compared with before intervention, PSQI also decreased significantly after intervention ($p = 0.002$).

As for RBDSQ, there was no significant difference between taVNS group and sham taVNS group ($p = 0.159$) before intervention. After intervention, there was also no significant difference between taVNS group and sham taVNS group ($p = 0.184$). However, in taVNS group, compared with before intervention, RBDSQ decreased significantly after intervention ($p = 0.025$). In sham taVNS group, compared with before intervention, there was also significant difference after intervention ($p = 0.005$).

Concerning ESS, there was no significant difference between taVNS group and sham taVNS group ($p = 0.837$) before intervention. After intervention, there was also no significant difference between taVNS group and sham taVNS group ($p = 0.712$). However, in taVNS group, compared with before intervention, ESS decreased significantly after intervention ($p < 0.001$). In sham taVNS group, compared with before intervention, ESS also decreased significantly after intervention ($p = 0.002$).

3.4.5. FAQ

About FAQ, there was no significant difference between taVNS group and sham taVNS group ($p = 0.431$) before intervention. After intervention, there was also no significant difference between taVNS group and sham taVNS group ($p = 0.907$). In taVNS group, compared with before intervention, FAQ reduced significantly after intervention ($p = 0.006$). In sham taVNS group, compared with before intervention, it also decreased significantly after intervention ($p = 0.000$).

3.5. Adverse reactions

In taVNS group, 1 patient with a history of tympanic membrane perforation reported mild toothache, sore throat and tinnitus after intervention for 13 weeks and was asked to stop intervention. Then this patient returned to normal after one week. There was no adverse reaction in sham taVNS group. During the research, physiological indicators of the patients in the two groups including blood pressure, respiration and heart rate, were normal. Dizziness, headache, skin lesions and other adverse reactions did not occur.

4. Discussion

This research demonstrated that taVNS is effective on improving MoCA-B, N5 and N7 in AVLT-H to various degrees. The targets of taVNS may be overall cognition, memory (including immediate recall and delayed recall).

This study showed that taVNS does not impact significantly on STTA when comparing the scores of STTA in taVNS group and sham taVNS after intervention, or when comparing the value differences of STTA before treatment and after treatment in both taVNS group

and sham taVNS group. Interestingly, when comparing the value differences of STTB before treatment and after treatment in both taVNS group and sham taVNS group, the value decreased significantly in both groups. The reason is maybe that STTB is a more complex task than STTA.

In this study, the scores of AFT at the baseline in taVNS group and sham taVNS group were significantly different. Therefore, whether taVNS impact on AFT still needs to be further investigated.

For STTA, STTB, BNT, PSQI, RBDSQ, ESS and FAQ, there were no statistical differences when comparing the values in taVNS group and sham taVNS group after 24 weeks of intervention, although there were significant differences when comparing the value difference of before intervention and after intervention in both taVNS and sham taVNS group. Whether taVNS impact on STTA, STTB, BNT, PSQI, RBDSQ, ESS and FAQ in a longer time and in a larger sample size still needs to be further investigated.

As for the effectiveness of taVNS on MoCA-B, a mean improvement of 3.24 points after 6 months of taVNS and a mean improvement of 0.33 points after 6 months of sham taVNS. As for the effectiveness of taVNS on N5 and N7, a mean improvement of 1.6 and 2.48 points respectively after 6 months of taVNS, whereas a mean improvement of 0.593 and 0.815 points after 6 months of sham taVNS. The result is similar to that of Sjögren et al. [11], who demonstrated that 7 out of 10 patients with AD were responders according to the ADAS-cog (median improvement of 3.0 points) after 3 months of VNS, and that 7 out of 10 patients with AD were responders according to the ADAS-cog (median improvement of 2.5 points) after 6 months in a longer intervention time, which corresponded to the minimal duration for evaluating cognitive functions.

Vagus nerve stimulation (VNS) can improve cognitive function. Sun et al. [10] have demonstrated that invasive VNS surgery can improve the cognitive function of patients with epilepsy. The cognitive performance of these patients with epilepsy, early visual event-related potential N1, and frontal alpha asymmetry were studied when cyclic vagus nerve stimulation was on and when it was off. VNS improved working memory performance as seen in reduced errors on a subtask that relied on working memory and increased N1 amplitude. Sjögren et al. [11] demonstrated that 9 out of 10 patients were responders according to the MMSE (median improvement of 1.5 points) after 3 months of treatment, and that 7 out of 10 patients were responders according to the MMSE (median improvement of 2.5 points) after 6 months of treatment. Merrill et al. [12] later found that 7 of 17 patients (41.2%) and 12 of 17 patients (70.6%) improved or did not decline from baseline on the ADAS-cog and MMSE after 1 year, respectively.

The surface of the auricle is the only cover area where the vagus nerve distributes in the body surface. There are auricular acupoints related to viscera and bowels reflexed in the concha. Therefore, our team originally developed transcutaneous electrical stimulation of auricular acupoints within the distribution of vagus nerve. It can improve cognitive functions in patients with epilepsy and depression. It can also improve the high confidence recognition memory in healthy subjects [35]. Llanos et al. has demonstrated that it enhances memory consolidation [36]. It can enhance the cognitive control ability [37–41] and associative memory ability in healthy subjects [42]. Xu G et al. [34] showed that auricular acupoint pressing pill combined with auricular acupoint massage can improve the activities of daily living and cognitive ability of MCI patients.

These findings and our clinical practice manifesting that auriculotherapy is effective on treating cognitive function encouraged us to formulate a hypothesis. The hypothesis is that electrical stimulation of auricular branch of vagus nerve can be used as a method to treat MCI.

MCI is the precursor of AD. It indicates mild impairment of cognitive function with independent daily activity. It is diagnosed on

basis of Jak/Bondi's criteria [19]. To include the suitable patients in our study, we excluded participants who had (1) nervous system diseases, including cerebrovascular disease, encephalitis, brain tumor, brain trauma, epilepsy, Parkinson's disease and other neurological diseases, which might lead to cognitive decline; or (2) metabolic diseases, including anemia, thyroid dysfunction, folic acid and Vitamin B12 deficiency, which might lead to cognitive decline; or (3) severe mental illness, such as severe depression, or history of gas poisoning; or (4) history of general anesthesia; or (5) dementia; or (6) acute or serious life-threatening illness; or (7) severe vision, hearing or language problems and could not complete neuropsychological assessment; or (8) had the scores of HAMD and HAMA ≥ 7 .

In the trial, taVNS, also including the pair of two auricular acupoints, namely heart (CO₁₅) and kidney (CO₁₀), was performed 5 days a week through patients treating themselves at home under the guidance of acupuncturists.

The design of the control group in clinical and animal studies confirming the efficacy of acupuncture treatment is mostly an issue. For this double blinded, randomized controlled study, another pair of the two auricular acupoints stimulated for sham taVNS group, namely elbow (SF₃) and shoulder (SF_{4,5}), which are located at the superior and inferior scapha with no distribution of vagus nerve. In order to make sure that both the acupuncturist and the patient were blinded, two pairs of carbon-impregnated silicone electrodes, one of which was disconnected from electrical output. Therefore, both the acupuncturist and the patient could not differentiate taVNS from sham taVNS.

As for the safety of taVNS, it interprets that a patient with a history of tympanic membrane perforation might suffer from mild toothache, sore throat and tinnitus after taVNS for 13 weeks, but the patient returned to normal when the patient stopped taVNS. It is safe for patients with MCI without the history of tympanic membrane perforation up to now.

The underlying mechanism of the efficacy of taVNS will be clarified separately through examinations of fMRI and EEG. The changes of the brain function during the resting state and the state of memory coding task [43] are designed for detection through fMRI, which will be presented in our future articles. The complexity of signals is evaluated through EEG. For patients with MCI (taVNS group and sham taVNS group), the examinations of fMRI and EEG were conducted before the interventions and 24 weeks after the interventions. For normal control group, the examinations of fMRI and EEG are conducted only once without any intervention.

The limitations of this trial are as follows. The sample size was relatively small. It is possible that the intervention time is not long enough to impact N5 and N7 to a better extent than this trial which demonstrated that there was still not significant difference between taVNS group and sham taVNS group after intervention, although it demonstrated that there was an increasing trend when comparing the scores of N7 in taVNS group and sham taVNS group after intervention ($p = 0.056$). Therefore, a year of intervention in a larger sample size is required for further investigation in the future.

5. Conclusion

In conclusion, this trial supported the efficacy and safety of taVNS in patients with MCI. It provides an important choice for improving cognitive functions and offers us a potential non-drug way to sustain or improve cognitive functions in patients with AD.

Ethics approval and consent to participate

The study protocol complied with the declaration of Helsinki and was approved by the Institutional Ethics Committee of Institute of Acupuncture and Moxibustion, China Academy of Chinese

Medical Sciences, Beijing, China. Informed written consent was obtained before the beginning of the baseline tests.

Availability of data and materials

Data sharing can be available to readers upon reasonable request by correspondence.

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CRediT authorship contribution statement

All authors contributed to the design of the trial. Lei Wang: Conceptualization, Methodology, trial registration, Writing-drafting, funding acquisition, performing. Jinling Zhang: patients screening, performing. Chunlei Guo took part in screening patients. Jiakai He did the statistical analysis. Shuai Zhang drew the figures and reviewed the manuscript. Yu Wang gave suggestion of the design of the trial, promoted the process of this research, polished figures and reviewed the manuscript. Zhao Yanan reviewed the manuscript and gave suggestions for revision. Li Liang, Junying Wang, Liwei Hou, Shaoyuan Li, Yifei Wang and Lixiao Hao: gave suggestion to the design of the trial and reviewed the manuscript. Yufeng Zhao contributed to the design of the trial, including randomization and blinding. Mozheng Wu reviewed the manuscript in English grammar and writing. Jiliang Fang: resources, project administration. Peijing Rong: Conceptualization, Methodology, Funding acquisition, Project administration, Writing-reviewing&editing. All authors have read and approved the publication of the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Abbreviations

Aβ	anti-amyloid beta
AD	Alzheimer's Disease
AFT	animal fluence test
AVLT-H	auditory verbal learning test-HuaShan version
BNT	Boston naming test
Concha	CO
EEG	electroencephalogram
fMRI	functional magnetic resonance imaging
HAMA	Hamilton anxiety scale
HAMD	Hamilton depression scale
MCI	mild cognitive impairment
MOCA-B	Montreal cognitive assessment-basic

iVNS	invasive vagus nerve stimulation
PSQI	Pittsburgh sleep quality index
SD	standard deviation
SF	scaphoid fossa
STT-A&B	shape trails test A&B
taVNS	transcutaneous auricular vagus nerve stimulation
TENS	transcutaneous electrical nerve stimulator
VNS	vagus nerve stimulation

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.brs.2022.09.003>.

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